

```

Last login: Wed Oct 7 15:23:24 on ttys000
[(base) matthieulbb@MacBook-Air-de-Matthieu-3 ~ % cd ~/Documents/"git cours"/esilv-marketpulse-g01-t3M
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % pwd
/Users/matthieulbb/Documents/git cours/esilv-marketpulse-g01-t3M
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % ls
config          labs            src
CONTRIBUTING.md  readiness      TEAM_TEMPLATE.md
data            README.md
evidence        requirements.txt
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % ls -la
total 16
drwxr-xr-x  15 matthieulbb  staff   480  7 oct.  14:33 .
drwxr-xr-x   5 matthieulbb  staff   160  7 oct.  15:23 ..
-rw-r--r--@  1 matthieulbb  staff  6148  7 oct.  14:37 .DS_Store
drwxr-xr-x  12 matthieulbb  staff   384  7 oct.  14:34 .git
-rw-r--r--   1 matthieulbb  staff   178  7 oct.  14:33 .gitignore
drwxr-xr-x   3 matthieulbb  staff    96  7 oct.  14:34 config
-rw-r--r--   1 matthieulbb  staff  2943  7 oct.  14:33 CONTRIBUTING.md
drwxr-xr-x   3 matthieulbb  staff    96  7 oct.  14:34 data
drwxr-xr-x   3 matthieulbb  staff    96  7 oct.  14:34 evidence
drwxr-xr-x   6 matthieulbb  staff   192  7 oct.  14:34 labs
drwxr-xr-x   3 matthieulbb  staff    96  7 oct.  14:34 readiness
-rw-r--r--   1 matthieulbb  staff 10270  7 oct.  14:33 README.md
-rw-r--r--   1 matthieulbb  staff   133  7 oct.  14:33 requirements.txt
drwxr-xr-x   3 matthieulbb  staff    96  7 oct.  14:34 src
-rw-r--r--   1 matthieulbb  staff  1446  7 oct.  14:33 TEAM_TEMPLATE.md
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % cd data/sample
pwd
ls
/Users/matthieulbb/Documents/git cours/esilv-marketpulse-g01-t3M/data/sample
bloomberg_reference_expected.json  instruments.json
bloomberg_reference_sample.json    prices.csv
[(base) matthieulbb@MacBook-Air-de-Matthieu-3 sample % cat data/sample/instruments.json
cat: data/sample/instruments.json: No such file or directory
(base) matthieulbb@MacBook-Air-de-Matthieu-3 sample % cd ../../
pwd
/Users/matthieulbb/Documents/git cours/esilv-marketpulse-g01-t3M
[(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}

```

```

(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % python --version
Python 3.12.7
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % git --version
git version 2.47.0
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % git add -A
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}

```

```
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % grep AAPL data/sample/prices.csv
grep SP500 data/sample/prices.csv
2026-09-01, AAPL, 249.20, 251.50, 248.00, 250.00, 38000000
2026-09-02, AAPL, 250.80, 252.80, 249.60, 251.30, 39500000
2026-09-03, AAPL, 249.60, 251.30, 248.40, 249.80, 41000000
2026-09-04, AAPL, 251.30, 253.60, 250.10, 252.10, 42500000
2026-09-08, AAPL, 251.90, 253.90, 250.70, 252.40, 44000000
2026-09-09, AAPL, 254.00, 255.70, 252.80, 254.20, 38000000
2026-09-10, AAPL, 254.30, 256.60, 253.10, 255.10, 39500000
2026-09-11, AAPL, 254.20, 256.20, 253.00, 254.70, 41000000
2026-09-14, AAPL, 256.60, 258.30, 255.40, 256.80, 42500000
2026-09-15, AAPL, 257.20, 259.50, 256.00, 258.00, 44000000
2026-09-16, AAPL, 256.70, 258.70, 255.50, 257.20, 38000000
2026-09-17, AAPL, 259.20, 260.90, 258.00, 259.40, 39500000
2026-09-18, AAPL, 259.30, 261.60, 258.10, 260.10, 41000000
2026-09-21, AAPL, 259.10, 261.10, 257.90, 259.60, 42500000
2026-09-22, AAPL, 261.10, 262.80, 259.90, 261.30, 44000000
2026-09-23, AAPL, 261.20, 263.50, 260.00, 262.00, 38000000
2026-09-24, AAPL, 263.00, 265.00, 261.80, 263.50, 39500000
2026-09-25, AAPL, 262.60, 264.30, 261.40, 262.80, 41000000
2026-09-28, AAPL, 263.30, 265.60, 262.10, 264.10, 42500000
2026-09-29, AAPL, 264.50, 266.50, 263.30, 265.00, 44000000
2026-09-30, AAPL, 266.00, 267.70, 264.80, 266.20, 38000000
2026-09-01, SP500, 6592.00, 6615.00, 6580.00, 6600.00, 0
2026-09-02, SP500, 6607.00, 6627.00, 6595.00, 6612.00, 0
2026-09-03, SP500, 6596.00, 6613.00, 6584.00, 6598.00, 0
2026-09-04, SP500, 6612.00, 6635.00, 6600.00, 6620.00, 0
2026-09-08, SP500, 6623.00, 6643.00, 6611.00, 6628.00, 0
2026-09-09, SP500, 6640.00, 6657.00, 6628.00, 6642.00, 0
2026-09-10, SP500, 6642.00, 6665.00, 6630.00, 6650.00, 0
2026-09-11, SP500, 6642.00, 6662.00, 6630.00, 6647.00, 0
2026-09-14, SP500, 6663.00, 6680.00, 6651.00, 6665.00, 0
2026-09-15, SP500, 6664.00, 6687.00, 6652.00, 6672.00, 0
2026-09-16, SP500, 6663.00, 6683.00, 6651.00, 6668.00, 0
2026-09-17, SP500, 6683.00, 6700.00, 6671.00, 6685.00, 0
2026-09-18, SP500, 6684.00, 6707.00, 6672.00, 6692.00, 0
2026-09-21, SP500, 6683.00, 6703.00, 6671.00, 6688.00, 0
2026-09-22, SP500, 6703.00, 6720.00, 6691.00, 6705.00, 0
2026-09-23, SP500, 6704.00, 6727.00, 6692.00, 6712.00, 0
2026-09-24, SP500, 6715.00, 6735.00, 6703.00, 6720.00, 0
2026-09-25, SP500, 6713.00, 6730.00, 6701.00, 6715.00, 0
2026-09-28, SP500, 6720.00, 6743.00, 6708.00, 6728.00, 0
2026-09-29, SP500, 6729.00, 6749.00, 6717.00, 6734.00, 0
2026-09-30, SP500, 6740.00, 6757.00, 6728.00, 6742.00, 0
```

```
((base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % python src/main.py
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
AAPL
Apple Inc.
USD
SP500
S&P 500
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
((base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01, AAPL, 249.20, 251.50, 248.00, 250.00, 38000000
2026-09-01, SP500, 6592.00, 6615.00, 6580.00, 6600.00, 0
2026-09-02, AAPL, 250.80, 252.80, 249.60, 251.30, 39500000
2026-09-02, SP500, 6607.00, 6627.00, 6595.00, 6612.00, 0
2026-09-03, AAPL, 249.60, 251.30, 248.40, 249.80, 41000000
2026-09-03, SP500, 6596.00, 6613.00, 6584.00, 6598.00, 0
2026-09-04, AAPL, 251.30, 253.60, 250.10, 252.10, 42500000
2026-09-04, SP500, 6612.00, 6635.00, 6600.00, 6620.00, 0
2026-09-08, AAPL, 251.90, 253.90, 250.70, 252.40, 44000000
```

```

def main():
    instruments = load_instruments()
    prices = load_prices()

    print(type(instrument))
    print(instruments.keys())

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    print(instrument["ticker"])
    print(instrument["name"])
    print(instrument["currency"])
    print(benchmark["ticker"])
    print(benchmark["name"])

```

```

(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % python src/main.py
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '380
250.00
<class 'str'>
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % python src/main.py
21
21
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21

```

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    ✨ print(type(prices))
    print(type(prices[0]))
    print(prices[0])
    print(prices[0]["close"])
    print(type(prices[0]["close"]))
```

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])
    print(len(instrument_prices))
    print(len(benchmark_prices))
```

```
[(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % python src/main.py
250.0
266.2
6600.0
6742.0
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
[(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % python src/main.py
=== MarketPulse ===

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
Instrument
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00
```

```

def get_first_close(prices):
    first_row = prices[0]
    return float(first_row["close"])

def get_last_close(prices):
    last_row = prices[-1]
    return float(last_row["close"])

def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices([prices, instrument["ticker"]])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    print(get_first_close(instrument_prices))
    print(get_last_close(instrument_prices))
    print(get_first_close(benchmark_prices))
    print(get_last_close(benchmark_prices))

```

```

[(base) matthieulbb@MacBook-Air-de-Matthieu-3 esilv-marketpulse-g01-t3M % python src/main.py
=== MarketPulse ===

```

```

Market configuration
Period: 1 month
Interval: Daily

```

```

Instrument
AAPL - Apple Inc.
Observations : 21
First close  : 250.00 USD
Last close   : 266.20 USD

```

```

Benchmark
SP500 - S&P 500
Observations : 21
First close  : 6600.00
Last close   : 6742.00

```

Users > matthieulbb > Documents > git cours > esilv-marketpulse-g01-t3M > src > main.py > main

```
1  from pathlib import Path
2  import csv
3  import json
4
5
6  DATA_DIR = Path("data/sample")
7
8  # These starter values mirror config/settings.yml.
9  # settings.yml is a human-readable configuration contract in the CORE.
10 # Parsing YAML is optional and is not required by the 18-hour lab sequence.
11 LOOKBACK_LABEL = "1 month"
12 INTERVAL_LABEL = "Daily"
13
14
15 def load_instruments():
16     with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
17         return json.load(file)
18
19
20 def load_prices():
21     with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
22         return list(csv.DictReader(file))
23
24
25 def filter_prices(prices, ticker):
26     return [row for row in prices if row["ticker"] == ticker]
27
28
29 def get_first_close(prices):
30     first_row = prices[0]
31     return float(first_row["close"])
32
33
34 def get_last_close(prices):
35     last_row = prices[-1]
36     return float(last_row["close"])
37
38
39 def display_market_summary(asset, prices, show_currency=True):
40     first_close = get_first_close(prices)
41     last_close = get_last_close(prices)
42
43     if show_currency:
44         unit = " " + asset["currency"]
45     else:
46         unit = ""
47
48     print(f"{asset['ticker']} - {asset['name']}")
49     print(f"Observations : {len(prices)}")
50     print(f"First close : {first_close:.2f}{unit}")
51     print(f>Last close : {last_close:.2f}{unit}")
52
53
54 def main():
55     instruments = load_instruments()
56     prices = load_prices()
57
58     instrument = instruments["instrument"]
59     benchmark = instruments["benchmark"]
60
61     instrument_prices = filter_prices(prices, instrument["ticker"])
62     benchmark_prices = filter_prices(prices, benchmark["ticker"])
63
64     print("=== MarketPulse ===")
65     print()
66     print("Market configuration")
67     print(f"Period : {LOOKBACK_LABEL}")
68     print(f"Interval : {INTERVAL_LABEL}")
69     print()
70     print("Instrument")
71     display_market_summary(instrument, instrument_prices)
72     print()
73     print("Benchmark")
74     display_market_summary(benchmark, benchmark_prices, show_currency=False)
75
76
77 if __name__ == "__main__":
78     main()
```